

Mutual Fund Performance Analysis using Python and Power BI

Course: Data Analytics

Project: Mutual Fund Performance Analysis Dashboard

Tools Used: Python, Excel, Power BI

1. Business Problem

Mutual fund investors often face difficulties in comparing thousands of available funds based on multiple financial metrics such as returns, risk, expense ratio, and fund size. Making investment decisions without proper analysis can lead to poor portfolio performance.

The objective of this project is to build an interactive analytical dashboard that helps investors compare mutual funds, identify top-performing funds, evaluate risk-adjusted returns, and make informed investment decisions.

2. Project Objectives

The primary objectives of this project are:

- Analyze mutual fund performance across different categories.
 - Compare funds using multiple return periods (1 Year, 3 Years, and 5 Years).
 - Evaluate investment risk using Beta and Sharpe Ratio.
 - Identify the best-performing mutual funds.
 - Study the relationship between risk and returns.
 - Provide interactive dashboards for business decision-making.
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3. Dataset Description

The dataset contains detailed information about various Indian mutual funds.

Important Columns

Column	Description
Scheme Name	Name of Mutual Fund
AMC Name	Asset Management Company

Column	Description
Category	Fund Category
Returns 1Y	One-Year Return
Returns 3Y	Three-Year Return
Returns 5Y	Five-Year Return
Expense Ratio	Annual Fund Expense
Sharpe Ratio	Risk-adjusted Return Measure
Beta	Market Risk Indicator
Fund Size	Assets Under Management (₹ Cr)
Risk Level	Investment Risk Category

4. Methodology

The project was completed in the following stages:

Step 1: Data Collection

Collected the mutual fund dataset containing performance metrics and fund details.

Step 2: Data Cleaning

Performed data cleaning using Python:

- Removed duplicate records.
- Handled missing values.
- Corrected inconsistent category names.
- Converted data types.
- Removed unwanted columns.

Step 3: Exploratory Data Analysis (EDA)

EDA was performed using:

- Pandas
- Matplotlib

- Seaborn

The analysis focused on:

- Return distributions
- Category comparisons
- Expense ratio analysis
- Risk analysis
- Fund size distribution

Step 4: Feature Engineering

Created additional fields including:

- Composite Score
- Rank
- Selected Return (Dynamic Measure)
- Average Return
- Average Sharpe Ratio

Step 5: Dashboard Development

Developed an interactive Power BI dashboard using:

- KPI Cards
- Bar Charts
- Scatter Plot
- Pie Chart
- Summary Table
- Dynamic Slicers
- Tooltips
- Drill-through
- Cross Filtering

5. Tools and Technologies

Tool	Purpose
Python	Data Cleaning & Analysis
Pandas	Data Manipulation
NumPy	Numerical Operations
Matplotlib	Data Visualization
Seaborn	Statistical Visualization
Excel	Data Validation
Power BI	Dashboard Development
DAX	Measures and KPIs

6. Dashboard Features

The dashboard contains the following visualizations.

KPI Cards

- Average Return
 - Average Expense Ratio
 - Average Sharpe Ratio
 - Total Fund Size
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Top 30 Mutual Funds

Displays the top 30 mutual funds based on the selected return period.

Features:

- Dynamic ranking
- Tooltips
- Drill-down
- Drill-through

Average Returns by Category

Compares average returns across categories.

Supports:

- Category hierarchy
 - Dynamic return period
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Risk (Beta) vs Return Scatter Plot

Analyzes the relationship between investment risk and returns.

Bubble Size:

- Fund Size
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Category Distribution

Pie chart showing the number of funds in each category.

Summary Table

Displays:

- Scheme Name
 - AMC
 - Category
 - Return
 - Expense Ratio
 - Sharpe Ratio
 - Beta
 - Rank
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Slicers

Interactive filters include:

- Category
 - Return Period
 - AMC
 - Risk Level
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7. Business Insights

Insight 1

Equity mutual funds generated the highest average returns compared to other categories.

Insight 2

Several top-performing funds achieved high returns while maintaining moderate expense ratios.

Insight 3

Funds with higher Sharpe Ratios generally delivered better risk-adjusted performance.

Insight 4

The scatter plot indicates that higher risk does not always guarantee higher returns.

Insight 5

Large fund size does not necessarily imply superior returns, emphasizing the need for comprehensive performance evaluation.

Insight 6

Long-term investment performance (5-Year Return) provides a better indication of consistent fund performance than short-term returns.

Insight 7

Interactive filtering enables investors to compare funds by category, AMC, and risk level, improving investment analysis.

8. Key Findings

- Equity funds outperform most categories in long-term returns.
 - Low expense ratios alone do not guarantee higher returns.
 - Sharpe Ratio is an effective metric for evaluating risk-adjusted performance.
 - Dynamic return-period analysis helps compare short-term and long-term investment strategies.
 - Investors should evaluate multiple financial metrics instead of relying on returns alone.
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9. Recommendations

Based on the analysis:

- Investors should prioritize funds with high Sharpe Ratios and consistent long-term returns.
 - Expense Ratio should be considered along with return performance.
 - Diversifying investments across categories can reduce portfolio risk.
 - Long-term returns provide a more reliable measure than short-term performance.
 - Investors should analyze both risk and return before making investment decisions.
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10. Dashboard Interactivity

The dashboard includes:

- Dynamic slicers
- Cross filtering
- Drill-through pages
- Drill-down hierarchy

- Interactive tooltips
- Dynamic DAX measures
- Top N filtering

These features improve user experience and enable deeper data exploration.

11. Limitations

- Analysis is based on historical data and does not guarantee future returns.
 - Market conditions may change over time.
 - External economic factors are not included in the analysis.
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12. Future Enhancements

Possible improvements include:

- Live mutual fund data integration using APIs.
 - Portfolio recommendation system.
 - Predictive return forecasting using Machine Learning.
 - Benchmark comparison against market indices.
 - Mobile-optimized dashboard.
 - AI-powered investment recommendations.
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13. Conclusion

This project successfully demonstrates how Python and Power BI can be combined to analyze mutual fund performance and support investment decision-making. The interactive dashboard enables users to compare funds across different return periods, evaluate risk-adjusted performance, and explore insights using dynamic filtering, drill-through, and cross-filtering features. The solution provides a user-friendly analytical platform that helps investors identify high-performing mutual funds while considering risk, expenses, and long-term consistency.

Dashboard Screenshots

